

SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to

solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS:2

Duration : 45 MINUTES

Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I LATISHA S (192524071) (Name / Reg No) certify that this submission is my original work

and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

SOLUTIONS

1. Smart Home Automation and Intelligent Agents

A smart home system uses AI to control lighting, temperature, and security based on sensor data, environmental conditions, and user preferences.

1. Simple Reflex Agent

It acts only on the current input using IF-THEN rules.

Examples:

- If motion is detected → Turn ON light.
- If temperature is high → Turn ON AC.
- If an intruder is detected → Activate alarm.

Advantage: Fast and simple.

Limitation: Cannot remember past events or learn user preferences.

2. Model-Based Agent

It maintains an internal model of the environment and uses both current and past information.

Example: It remembers that the user is away and keeps lights and AC OFF to save energy.

Advantage: Handles incomplete information.

Limitation: More complex than a reflex agent.

3. Goal-Based Agent

It selects actions to achieve specific goals.

Goals:

- Maintain comfortable temperature.
- Save energy.
- Maintain home security.

Advantage: Can plan actions to achieve goals.

Limitation: May not select the most efficient option among several choices.

4. Utility-Based Agent

It selects the action that provides the best overall benefit by considering comfort, energy consumption, and security.

For example, it may choose a moderate AC temperature that provides comfort while reducing electricity usage.

Advantage: Can balance multiple objectives.

Limitation: Requires a suitable utility function.

Best Approach

A hybrid agent combining model-based, goal-based, and utility-based features is best for a smart home.

- Reflex: Handles emergencies quickly.
- Model-based: Maintains information about the home.
- Goal-based: Achieves comfort and security goals.
- Utility-based: Balances comfort, energy efficiency, and security.
- Learning: Learns user preferences over time.

Conclusion

Hybrid Intelligent Agent = Best for Smart Home Automation

It provides comfort, energy efficiency, security, and personalized control.

2. Robot in an Unknown Maze – UCS and IDS

A robot in an unknown maze must explore the environment and find the exit without knowing the maze layout beforehand. Two uninformed search methods are Uniform Cost Search (UCS) and Iterative Deepening Search (IDS).

1. Uniform Cost Search (UCS)

UCS expands the node with the lowest path cost using a priority queue.

$$f(n) = g(n)$$

where $g(n)$ is the cost from the start node.

Advantages:

- Finds the minimum-cost path.
- Works well when different movements have different costs.
- Complete when action costs are positive.

Disadvantages:

- Requires high memory.
- Can be slow because it may explore many nodes.

Complexity:

$$Time \approx O(b^{1+\lceil C^*/\epsilon \rceil})$$

$$Space \approx O(b^{1+\lceil C^*/\epsilon \rceil})$$

2. Iterative Deepening Search (IDS)

IDS repeatedly performs depth-limited DFS with increasing depth:

$$0, 1, 2, 3, \dots$$

It continues until the exit is found.

Advantages:

- Uses less memory.
- Complete for finite branching.
- Finds the shallowest solution when all actions have equal cost.

Disadvantages:

- Repeats searches at previous depths.
- Not suitable for different path costs.

Complexity:

$$Time = O(b^d)$$

$$Space = O(bd)$$

where b is the branching factor and d is the solution depth.

Intelligent Agents

- Simple Reflex Agent: Reacts only to current obstacles. It may get stuck or repeat actions.
- Model-Based Agent: Remembers explored areas, obstacles, and dead ends. This is better for an unknown maze.
- Goal-Based Agent: Has the goal of reaching the exit and searches for a suitable path.
- Utility-Based Agent: Chooses the best path based on factors such as cost, safety, and energy.

Best Approach

A Model-Based + Goal-Based Agent is most suitable because the robot needs to remember the maze and has a clear goal of reaching the exit.

- If all moves have equal cost and memory is limited → IDS is suitable.
- If movement costs are different → UCS is better because it finds the minimum-cost path.

Conclusion

Model-Based + Goal-Based Agent + UCS

is the best general choice when path costs vary, while IDS is a good choice for equal-cost movements with limited memory.